

Hofstadter physics in continuum model

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1 Lattice in magnetic field

For a general 2D lattice with lattice basis vectors $\mathbf{a}_1, \mathbf{a}_2$ (we set $\mathbf{a}_2 = a_2 \mathbf{e}_y$ and $a_{1x} > 0$) and reciprocal basis vectors $\mathbf{b}_1, \mathbf{b}_2$ ($\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$), the Hamiltonian can be written as

$$H(\mathbf{p}, \mathbf{r}) = T(\mathbf{p}) + V(\mathbf{r}), \quad (1)$$

where $T(\mathbf{p})$ is the kinetic energy, and $V(\mathbf{r})$ is the periodic potential satisfying $V(\mathbf{r}) = V(\mathbf{r} + n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2)$ for any integers n_1, n_2 . The Hamiltonian has the lattice translation symmetry $[H(\mathbf{p}, \mathbf{r}), e^{-i\mathbf{p} \cdot \mathbf{a}_i/\hbar}] = 0$.

Now under magnetic field $\mathbf{B} = \chi B \mathbf{e}_z$, the Hamiltonian becomes

$$H(\boldsymbol{\pi}, \mathbf{r}) = T(\boldsymbol{\pi}) + V(\mathbf{r}), \quad (2)$$

where $\mathbf{A}$ is the vector potential, $\boldsymbol{\pi} = \mathbf{p} + e\mathbf{A}$. In calculations we choose the Landau gauge: $\mathbf{A} = \chi B x \mathbf{e}_y$. The vector potential breaks the usual translation symmetry, but we still can find two magnetic translation operators $t(\mathbf{a}_i)$ so that $[H(\boldsymbol{\pi}, \mathbf{r}), t(\mathbf{a}_i)] = 0$. The two operators are

$$t(\mathbf{a}_1) = e^{-i\boldsymbol{\pi}' \cdot \mathbf{a}_1/\hbar}, \quad t(\mathbf{a}_2) = e^{-i\boldsymbol{\pi}' \cdot \mathbf{a}_2/\hbar} = e^{-i\mathbf{p} \cdot \mathbf{a}_2/\hbar}. \quad (3)$$

where $\boldsymbol{\pi}' = \boldsymbol{\pi} - e\mathbf{B} \times \mathbf{r}$. Although $t(\mathbf{a}_i)$ commute with H , they do not commute with each other: $t(\mathbf{a}_1)t(\mathbf{a}_2) = \exp(-i2\pi\Phi/\Phi_0)t(\mathbf{a}_2)t(\mathbf{a}_1)$, where $\Phi = \chi B a_{1x} a_2$ is the (signed) magnetic flux in a unit cell and $\Phi_0 = h/e$ is the flux quantum. We consider the case $|\Phi|/\Phi_0 = p/q$, where p and q are co-prime integers, we then have

$$[t^q(\mathbf{a}_1), t(\mathbf{a}_2)] = 0. \quad (4)$$

So the Hilbert space of H can be classified according to eigenvalues of $t^q(\mathbf{a}_1)$ and $t(\mathbf{a}_2)$.

2 Landau levels as basis

If $V(\mathbf{r})$ is absent, the problem is the traditional Landau level (LL) problem. In this case we only need to diagonalize $T(\boldsymbol{\pi})$. The torus system can be viewed as a magnetic lattice with $N_\phi = N_1 \times N_2$ sites ($N_\phi = A/(2\pi l_B^2)$ is the LL degeneracy) and lattice vectors $\mathbf{m}_1, \mathbf{m}_2$ ($\mathbf{m}_2 = m_2 \mathbf{e}_y$, $m_{1x} > 0$). The Bloch LLs are (see the note ‘‘LL formulation’’)

$$\begin{aligned} |\psi_{n,\mathbf{k}}\rangle &= \frac{e^{i\chi\pi k_2^2 m_{1y}/m_2}}{\sqrt{N_1 N_t}} \sum_{s=0}^{N_1-1} \sum_{l \in \mathbb{Z}} e^{i2\pi(s+lN_1)k_1} t^{s+lN_1}(\mathbf{m}_1) |\varphi_{n, \frac{2\pi}{m_2} k_2}\rangle \\ &= \frac{1}{\sqrt{N_1 N_t}} \sum_{s \in \mathbb{Z}} e^{is2\pi k_1} e^{i\chi\pi \frac{m_{1y}}{m_2} (k_2 - \chi s)^2} |\varphi_{n, \frac{2\pi}{m_2} (k_2 - \chi s)}\rangle, \end{aligned} \quad (5)$$

where $\mathbf{k} = k_1 \mathbf{g}_1 + k_2 \mathbf{g}_2 \in \text{MBZ}$ (magnetic Brillouin zone, $k_1, k_2 \in [0, 1)$), $|\varphi\rangle$ are LLs on the cylinder, and $\mathbf{g}_i$ are magnetic reciprocal vectors ($\mathbf{g}_i \cdot \mathbf{m}_j = 2\pi \delta_{ij}$). Each magnetic unit cell has area $2\pi l_B^2$. The Bloch LLs are eigenstates of magnetic translations

$$t(\mathbf{m}_1)|\psi_{n,\mathbf{k}}\rangle = e^{-i2\pi k_1} |\psi_{n,\mathbf{k}}\rangle, \quad t(\mathbf{m}_2)|\psi_{n,\mathbf{k}}\rangle = e^{-i2\pi k_2} |\psi_{n,\mathbf{k}}\rangle. \quad (6)$$

The plane wave matrix element under such basis is ($\mathbf{q} = q_1 \mathbf{g}_1 + q_2 \mathbf{g}_2$)

$$\langle n', \mathbf{k}' | e^{i\mathbf{q} \cdot \mathbf{r}} | n, \mathbf{k} \rangle = \delta_{k'_1, [k_1+q_1]} \delta_{k'_2, [k_2+q_2]} e^{-i\chi 2\pi k'_1 (k'_2 - k_2 - q_2)} e^{-i\chi 2\pi q_1 (k_2 + \frac{q_2}{2})} F_{n'n} \left(\frac{q_x + i\chi q_y l_B}{\sqrt{2}} \right), \quad (7)$$

where $[k_i]$ denotes k_i reduced modulo 1 into $[0, 1)$, and (L_n^α) is the associated Laguerre polynomial

$$F_{n'n}(Q) = \begin{cases} e^{-\frac{|Q|^2}{2}} \sqrt{\frac{n!}{n'}} (iQ)^{n-n'} L_{n'-n'}^{n-n'}(|Q|^2) & (n' \leq n) \\ e^{-\frac{|Q|^2}{2}} \sqrt{\frac{n!}{n'}} (iQ^*)^{n'-n} L_{n'-n}^{n'-n}(|Q|^2) & (n' \geq n) \end{cases}. \quad (8)$$

In the case of $\Phi/\Phi_0 = \chi p/q$, i.e., $a_{1x} a_2 = 2\pi l_B^2 p/q$, the magnetic and lattice cells are commensurate: $p\mathbf{m}_1 \times \mathbf{m}_2 = q\mathbf{a}_1 \times \mathbf{a}_2$, so we set the magnetic lattice $\mathbf{m}_1 = \frac{q}{p} \mathbf{a}_1$, $\mathbf{m}_2 = \mathbf{a}_2$ and $\mathbf{g}_1 = \frac{p}{q} \mathbf{b}_1$, $\mathbf{g}_2 = \mathbf{b}_2$. For convenience, we define the commensurate supercell lattice vectors $\mathbf{L}_{1,2}$ and reciprocal lattice vectors $\mathbf{P}_{1,2}$ as

$$\mathbf{L}_1 = q\mathbf{a}_1 = p\mathbf{m}_1, \quad \mathbf{L}_2 = \mathbf{a}_2 = \mathbf{m}_2, \quad (9)$$

$$\mathbf{P}_1 = \mathbf{b}_1/q = \mathbf{g}_1/p, \quad \mathbf{P}_2 = \mathbf{b}_2 = \mathbf{g}_2, \quad (10)$$

satisfying $\mathbf{L}_i \cdot \mathbf{P}_j = 2\pi \delta_{ij}$. Then all the momentum $\mathbf{k}$ in the supercell Brillouin zone (SBZ) can be parametrized as $\mathbf{k} = k_1 \mathbf{P}_1 + k_2 \mathbf{P}_2$ with $k_1, k_2 \in [0, 1)$. Since the Hamiltonian commutes with magnetic translations $t(\mathbf{L}_1) = t^q(\mathbf{a}_1)$ and $t(\mathbf{L}_2) = t(\mathbf{a}_2)$, we can use their common eigenstates to diagonalize the Hamiltonian.

One should be very careful that, it is the LLs in the MBZ that span the whole Hilbert space so we should use LL basis within full MBZ, which is p times larger than the SBZ (the lattice BZ, on the other hand, is q times larger than the SBZ). The momentum $\tilde{\mathbf{k}} \in \text{MBZ}$ can be parameterized as $\tilde{\mathbf{k}} = \mathbf{k} + r\mathbf{P}_1 = (k_1 + r)\mathbf{P}_1 + k_2 \mathbf{P}_2 = \frac{k_1+r}{p} \mathbf{g}_1 + k_2 \mathbf{g}_2$, with $\mathbf{k} \in \text{SBZ}$, and $r = 0, 1, \dots, p-1$. We thus relabel Eq. (5) as

$$|n, r, \mathbf{k}\rangle = |\psi_{n, \mathbf{k}+r\mathbf{P}_1}\rangle = |\psi_{n, \frac{k_1+r}{p} \mathbf{g}_1 + k_2 \mathbf{g}_2}\rangle = \frac{1}{\sqrt{N_1 N_t}} \sum_{s \in \mathbb{Z}} e^{is2\pi \frac{k_1+r}{p}} e^{i\chi\pi \frac{q}{p} \frac{a_{1y}}{a_2} (k_2 - \chi s)^2} |\varphi_{n, \frac{2\pi}{a_2} (k_2 - \chi s)}\rangle. \quad (11)$$

We can check $|n, r+p, \mathbf{k}\rangle = |n, r, \mathbf{k}\rangle$. Using Eq. (6), we have

$$t(\mathbf{L}_1)|n, r, \mathbf{k}\rangle = t^p(\mathbf{m}_1)|n, r, \mathbf{k}\rangle = e^{-i2\pi k_1} |n, r, \mathbf{k}\rangle, \quad t(\mathbf{L}_2)|n, r, \mathbf{k}\rangle = t(\mathbf{m}_2)|n, r, \mathbf{k}\rangle = e^{-i2\pi k_2} |n, r, \mathbf{k}\rangle, \quad (12)$$

so Eq. (11) are suitable to diagonalize H (the orthonormality follows easily from Eq. (5): $\langle n', r', \mathbf{k}' | n, r, \mathbf{k} \rangle = \delta_{n', n} \delta_{r', r} \delta_{\mathbf{k}', \mathbf{k}}$). The following target is to determine the matrix elements in these basis.

3 Matrix element

3.1 Kinetic energy terms

We calculate the Hamiltonian matrix elements in basis (11). The kinetic energy is simple since the involved operator π is related to the ladder operators

$$\pi_x = i \frac{\hbar}{\sqrt{2}l_B} (a^\dagger - a), \quad \pi_y = \chi \frac{\hbar}{\sqrt{2}l_B} (a^\dagger + a), \quad (13)$$

which act only on the LL energy index n , and is diagonal in $\mathbf{k}$ and r . The results are

$$\langle n' r' \mathbf{k} | a | n r \mathbf{k} \rangle = \sqrt{n} \delta_{n'+1, n} \delta_{r' r}, \quad \langle n' r' \mathbf{k} | a^\dagger | n r \mathbf{k} \rangle = \sqrt{n'} \delta_{n', n+1} \delta_{r' r}, \quad \langle n' r' \mathbf{k} | a^\dagger a | n r \mathbf{k} \rangle = n \delta_{n' n} \delta_{r' r}. \quad (14)$$

We notice that the kinetic energy terms are independent on $\mathbf{k}$, which is consistent with the fact that the Landau levels of electron gas with purely kinetic energies (and without potentials) are completely flat.

3.2 Potential energy terms

For the lattice potential $V(\mathbf{r})$, in general it can be expanded as plane waves

$$V(\mathbf{r}) = \sum_{\mathbf{G}} V_{\mathbf{G}} e^{i\mathbf{G} \cdot \mathbf{r}}, \quad \mathbf{G} = G_1 \mathbf{b}_1 + G_2 \mathbf{b}_2 = \frac{q}{p} G_1 \mathbf{g}_1 + G_2 \mathbf{g}_2, \quad (15)$$

where G_i are integers. So we only need the plane wave matrix elements. Rewriting Eq. (7), we get (please be careful that in Eq. (7) q_1, q_2 are defined as coefficients of $\mathbf{G}$ in the vector basis of $\mathbf{g}_1, \mathbf{g}_2$, which are $\frac{q}{p} G_1$ and G_2 here)

$$\begin{aligned} \langle n' r' \mathbf{k}' | e^{i\mathbf{G} \cdot \mathbf{r}} | n r \mathbf{k} \rangle &= F_{n'n} \left(\frac{G_x + i\chi G_y}{\sqrt{2}} l_B \right) \delta_{\frac{k'_1+r'}{p}, [\frac{k_1+r+G_1 q}{p}]} \delta_{k'_2, [k_2+G_2]} e^{-i\chi 2\pi \frac{k'_1+r'}{p} (k'_2 - k_2 - G_2)} e^{-i\chi 2\pi \frac{q}{p} G_1 (k_2 + \frac{G_2}{2})} \\ &= F_{n'n} \left(\frac{G_x + i\chi G_y}{\sqrt{2}} l_B \right) \delta_{k'_1 k_1} \delta_{k'_2 k_2} \delta_{\frac{r'}{p}, [\frac{r+G_1 q}{p}]} e^{i\chi 2\pi \frac{k_1+r'}{p} G_2} e^{-i\chi 2\pi \frac{q}{p} G_1 (k_2 + \frac{G_2}{2})} \\ &= F_{n'n} \left(\frac{G_x + i\chi G_y}{\sqrt{2}} l_B \right) \delta_{k'_1 k_1} \delta_{k'_2 k_2} \sum_{s \in \mathbb{Z}} \delta_{r', r+G_1 q + s p} e^{i\chi 2\pi \frac{k_1+r+G_1 q}{p} G_2} e^{-i\chi 2\pi \frac{q}{p} G_1 (k_2 + \frac{G_2}{2})} \\ &= F_{n'n} \left(\frac{G_x + i\chi G_y}{\sqrt{2}} l_B \right) \delta_{k'_1 k_1} \delta_{k'_2 k_2} e^{i\chi 2\pi \frac{k_1+r}{p} G_2 - i\chi 2\pi \frac{q}{p} G_1 (k_2 - \frac{G_2}{2})} \delta_{r', r+G_1 q}^{\text{mod } p}. \end{aligned} \quad (16)$$

For each (r', r) , at most one s survives in the summation. Eq. (17) is dense in n , but sparse in r . In calculations we make a cutoff, i.e., we truncate N_{LL} LLs, corresponding to an energy window $\sim \hbar^2 N_{\text{LL}} / (m l_B^2)$. The computation cost increases when $B \rightarrow 0$.

To conveniently calculate $F_{n'n}(Q)$, we rewrite Eq. (8) as (suppose $|Q| > 0$)

$$F_{n'n}(Q) = \begin{cases} e^{i(\frac{\pi}{2} + \theta_Q)(n-n')} \exp \left[\frac{\ln \Gamma(n'+1) - \ln \Gamma(n+1)}{2} - \frac{|Q|^2}{2} + (n-n') \ln |Q| \right] L_{n'}^{n-n'}(|Q|^2) & (n' \leq n) \\ e^{i(\frac{\pi}{2} - \theta_Q)(n'-n)} \exp \left[\frac{\ln \Gamma(n+1) - \ln \Gamma(n'+1)}{2} - \frac{|Q|^2}{2} + (n'-n) \ln |Q| \right] L_n^{n'-n}(|Q|^2) & (n' \geq n) \end{cases}, \quad (17)$$

where $\theta_Q = \arg(Q)$, $\Gamma(n)$ is the gamma function, and for the Laguerre polynomials we recursively use

$$L_{n+1}^\alpha(x) = \frac{(2n+1+\alpha-x)L_n^\alpha(x) - (n+\alpha)L_{n-1}^\alpha(x)}{n+1}, \quad (18)$$

starting from $L_0^\alpha(x) = 1$, $L_1^\alpha(x) = 1 + \alpha - x$.

3.3 Properties of Hofstadter bands

Now we list the properties of the Hofstadter bands calculated using the above method.

We prove that the spectrum of H is q -fold degenerate when $q > 1$. Suppose $H|\phi_{\nu, \mathbf{k}}\rangle = E_{\nu, \mathbf{k}}|\phi_{\nu, \mathbf{k}}\rangle$ where $|\phi_{\nu, \mathbf{k}}\rangle$ is the eigenstate of H with $t(\mathbf{L}_1)|\phi_{\nu, \mathbf{k}}\rangle = e^{-i2\pi k_1}|\phi_{\nu, \mathbf{k}}\rangle$ and $t(\mathbf{L}_2)|\phi_{\nu, \mathbf{k}}\rangle = e^{-i2\pi k_2}|\phi_{\nu, \mathbf{k}}\rangle$. Then

$t(\mathbf{a}_1)|\phi_{\nu,\mathbf{k}}\rangle$ is also an eigenstate of H with the same eigenvalue, because $[H, t(\mathbf{a}_1)] = 0$. In addition, using that $t(\mathbf{a}_1)t(\mathbf{a}_2) = e^{-i\chi 2\pi p/q} t(\mathbf{a}_2)t(\mathbf{a}_1)$, we can prove $t(\mathbf{L}_1)t(\mathbf{a}_1)|\phi_{\nu,\mathbf{k}}\rangle = e^{-i2\pi k_1} t(\mathbf{a}_1)|\phi_{\nu,\mathbf{k}}\rangle$ and $t(\mathbf{L}_2)t(\mathbf{a}_1)|\phi_{\nu,\mathbf{k}}\rangle = e^{i\chi 2\pi p/q} t(\mathbf{a}_1)t(\mathbf{a}_2)|\phi_{\nu,\mathbf{k}}\rangle = e^{-i2\pi(k_2 - \chi p/q)} t(\mathbf{a}_1)|\phi_{\nu,\mathbf{k}}\rangle$, so $t(\mathbf{a}_1)|\phi_{\nu,\mathbf{k}}\rangle$ is an eigenstate of H with the $t(\mathbf{L}_1)$ eigenvalue $e^{-i2\pi k_1}$ and $t(\mathbf{L}_2)$ eigenvalue $e^{-i2\pi(k_2 - \chi p/q)}$. Please be careful that $[k_2 - \chi p/q] \neq k_2$ since p, q are coprime. So $t(\mathbf{a}_1)|\phi_{\nu,\mathbf{k}}\rangle$ is indeed different from $|\phi_{\nu,\mathbf{k}}\rangle$ due to its different $t(\mathbf{L}_2)$ eigenvalue. More generally, we can prove $t^s(\mathbf{a}_1)|\phi_{\nu,\mathbf{k}}\rangle$ with $s = 1, 2, \dots, q-1$ are all degenerate with $|\phi_{\nu,\mathbf{k}}\rangle$ with the same $t(\mathbf{L}_1)$ eigenvalue $e^{-i2\pi k_1}$ but with different $t(\mathbf{L}_2)$ eigenvalues $e^{-i2\pi(k_2 - s\chi p/q)}$. This proves the q -fold degeneracy of the Hofstadter bands. Explicitly, for the LL basis Eq. (11), we can prove

$$t(\mathbf{a}_1)|n, r, k_1, k_2\rangle = |n, r, k_1, k_2 - \chi p/q\rangle = e^{i\chi 2\pi N \frac{k_1+r}{p}} |n, r, k_1, [k_2 - \chi p/q]\rangle, \quad (19)$$

where $N = k_2 - \chi p/q - [k_2 - \chi p/q]$ is some integer.

4 Examples

4.1 Square lattice

For square lattice, $\mathbf{a}_1 = (a, 0)$, $\mathbf{a}_2 = (0, a)$, the Hamiltonian reads ($V_0 > 0$)

$$H(\boldsymbol{\pi}, \mathbf{r}) = \frac{\boldsymbol{\pi}^2}{2m} + V(\mathbf{r}) = \frac{\hbar^2}{l_B^2 m} \left(a^\dagger a + \frac{1}{2} \right) - V_0 (e^{i\mathbf{b}_1 \cdot \mathbf{r}} + e^{i\mathbf{b}_2 \cdot \mathbf{r}} + e^{-i\mathbf{b}_1 \cdot \mathbf{r}} + e^{-i\mathbf{b}_2 \cdot \mathbf{r}}), \quad (20)$$

the matrix elements are ($b = 2\pi/a$)

$$\begin{aligned} \langle n' r' \mathbf{k} | \frac{\hbar^2}{ml_B^2} \left(a^\dagger a + \frac{1}{2} \right) | nr \mathbf{k} \rangle &= \frac{\hbar^2}{ml_B^2} \left(n + \frac{1}{2} \right) \delta_{n'n} \delta_{r'r}, \\ \langle n' r' \mathbf{k} | e^{i\mathbf{b}_1 \cdot \mathbf{r}} | nr \mathbf{k} \rangle &= F_{n'n} \left(\frac{bl_B}{\sqrt{2}} \right) e^{-i\chi 2\pi \frac{q}{p} k_2} \delta_{r', r+q}^{\text{mod } p}, \\ \langle n' r' \mathbf{k} | e^{i\mathbf{b}_2 \cdot \mathbf{r}} | nr \mathbf{k} \rangle &= F_{n'n} \left(i\chi \frac{bl_B}{\sqrt{2}} \right) e^{i\chi 2\pi \frac{k_1+r}{p}} \delta_{r', r}. \end{aligned} \quad (21)$$

4.2 Triangle lattice

The lattice vectors $\mathbf{a}_1 = a \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right)$, $\mathbf{a}_2 = a(0, 1)$, $\mathbf{G}_1 = \mathbf{b}_1 = \frac{4\pi}{\sqrt{3}a} (1, 0)$, $\mathbf{G}_2 = \mathbf{b}_2 = \frac{4\pi}{\sqrt{3}a} \left(-\frac{1}{2}, \frac{\sqrt{3}}{2} \right)$. The kinetic energy is the same as the square-lattice one, while the potential term becomes ($\mathbf{G}_3 = -\mathbf{b}_1 - \mathbf{b}_2$)

$$V(\mathbf{r}) = -V_0 [e^{i\mathbf{G}_1 \cdot \mathbf{r}} + e^{i\mathbf{G}_2 \cdot \mathbf{r}} + e^{i\mathbf{G}_3 \cdot \mathbf{r}} + e^{-i\mathbf{G}_1 \cdot \mathbf{r}} + e^{-i\mathbf{G}_2 \cdot \mathbf{r}} + e^{-i\mathbf{G}_3 \cdot \mathbf{r}}]. \quad (22)$$

The matrix elements are

$$\begin{aligned} \langle n' r' \mathbf{k} | e^{i\mathbf{G}_1 \cdot \mathbf{r}} | nr \mathbf{k} \rangle &= F_{n'n} \left(\frac{G_{1x} + i\chi G_{1y}}{\sqrt{2}} l_B \right) e^{-i\chi 2\pi \frac{q}{p} k_2} \delta_{r', r+q}^{\text{mod } p}, \\ \langle n' r' \mathbf{k} | e^{i\mathbf{G}_2 \cdot \mathbf{r}} | nr \mathbf{k} \rangle &= F_{n'n} \left(\frac{G_{2x} + i\chi G_{2y}}{\sqrt{2}} l_B \right) e^{i\chi 2\pi \frac{k_1+r}{p}} \delta_{r', r}, \\ \langle n' r' \mathbf{k} | e^{i\mathbf{G}_3 \cdot \mathbf{r}} | nr \mathbf{k} \rangle &= F_{n'n} \left(\frac{G_{3x} + i\chi G_{3y}}{\sqrt{2}} l_B \right) e^{-i\chi 2\pi \frac{k_1+r}{p} + i\chi 2\pi \frac{q}{p} (k_2 + \frac{1}{2})} \delta_{r', r-q}^{\text{mod } p}. \end{aligned} \quad (23)$$

4.3 Twisted bilayer graphene

Twisted bilayer graphene (TBG) has the continuum model, so under magnetic field it can also be diagonalized using the above method. We leave this part in another note called “Note_TB_G_Hofstadter”.

5 The Diophantine equation

For simple lattice with flux $\Phi/\Phi_0 = \chi p/q$, where p and q are co-prime, the energy spectrum splits into a set of Hofstadter bands. The ν -th gap (from low- to high-energy) is characterized by a pair of integers (t_ν, s_ν),

which satisfies the Diophantine equation ($\chi = B_z/B = \pm 1$)

$$\nu = \chi t_\nu p + s_\nu q. \quad (24)$$

Specifically, t_ν is the Chern number of the ν -th gap (summation of all the Chern numbers of bands below the ν -th gap), i.e., the Chern number of the ν -th band is $\sigma_\nu = t_\nu - t_{\nu-1}$.

5.1 A band counting proof

Consider the bands that have been filled below the ν -th gap. Suppose the ν -th gap does not close when we change the magnetic field strength. Let ρ_ν be the electron density of the system (electron number per lattice unit cell). For magnetic flux $\Phi/\Phi_0 = \chi p/q$, each unit supercell contains q lattice unit cells. Therefore,

$$\rho_\nu = \nu/q. \quad (25)$$

On the other hand, from the Streda formula ($A_0 = a_{1x}a_2$ is the lattice unit cell area)

$$\sigma_{xy} = e \frac{\partial n}{\partial B_z} \implies \frac{\partial \rho_\nu}{\partial(\Phi/\Phi_0)} = \Phi_0 \frac{\partial(A_0 n)}{\partial(A_0 B_z)} = \frac{h}{e^2} \sigma_{xy}. \quad (26)$$

Using the quantized Hall conductivity $\sigma_{xy} = \frac{e^2}{h} t_\nu$ ($t_\nu \in \mathbb{Z}$), then ρ_ν as a function of Φ can only change according to

$$\rho_\nu = t_\nu \Phi/\Phi_0 + s_\nu, \quad t_\nu, s_\nu \in \mathbb{Z}. \quad (27)$$

where s_ν is another integer corresponding to the zero-flux electron density (we need track the gap that keeps open when we decrease $|\Phi|/\Phi_0$ from a finite value to zero, which usually arrives at an intrinsic zero-field gap opened purely by the lattice potential, and hence s_ν is an integer). The integrality of s_ν is the gap-labeling condition; equivalently, it follows from the boundary-condition proof below. Putting Eq. (25) and $\Phi/\Phi_0 = \chi p/q$ into the above equation, we reach the Diophantine equation

$$\frac{\nu}{q} = t_\nu \chi \frac{p}{q} + s_\nu \implies \nu = \chi t_\nu p + s_\nu q. \quad (28)$$

5.2 A boundary-condition proof

We now give a boundary-condition proof of the Diophantine equation. The proof is adapted to the magnetic-translation convention used above, where the commuting magnetic translations are $t(\mathbf{L}_1) = t^q(\mathbf{a}_1)$ and $t(\mathbf{L}_2) = t(\mathbf{a}_2)$. Thus a magnetic Bloch eigenstate satisfies $t(\mathbf{L}_1)|\phi_{\mu,k_1,k_2}\rangle = e^{-i2\pi k_1}|\phi_{\mu,k_1,k_2}\rangle$ and $t(\mathbf{L}_2)|\phi_{\mu,k_1,k_2}\rangle = e^{-i2\pi k_2}|\phi_{\mu,k_1,k_2}\rangle$, where μ labels an isolated Hofstadter band, and $k_1, k_2 \in [0, 1)$. According to the discussion in section 3.3, we know that, up to a phase:

$$t(\mathbf{a}_1)|\phi_{\mu,k_1,k_2}\rangle = e^{i\theta_\mu(k_1,k_2)}|\phi_{\mu,k_1,k_2-\chi p/q}\rangle. \quad (29)$$

For an isolated band with Chern number σ_μ , we choose the boundary condition

$$|\phi_{\mu,k_1+1,k_2}\rangle = |\phi_{\mu,k_1,k_2}\rangle, \quad |\phi_{\mu,k_1,k_2+1}\rangle = e^{i2\pi\sigma_\mu k_1}|\phi_{\mu,k_1,k_2}\rangle. \quad (30)$$

This boundary condition carries Chern number σ_μ , since the transition function across the k_2 -boundary has winding number σ_μ along the k_1 direction.

Using the remaining gauge freedom, the phase in the magnetic-translation relation may be chosen as

$$t(\mathbf{a}_1)|\phi_{\mu,k_1,k_2}\rangle = e^{i2\pi m_\mu k_1}|\phi_{\mu,k_1,k_2-\chi p/q}\rangle. \quad (31)$$

Here m_μ must be an integer. Indeed, because the state is periodic under $k_1 \rightarrow k_1 + 1$, consistency of Eq. (31) gives $e^{i2\pi m_\mu} = 1$, and therefore $m_\mu \in \mathbb{Z}$.

Now apply $t(\mathbf{a}_1)$ q times. Since $t^q(\mathbf{a}_1) = t(\mathbf{L}_1)$, Eq. (31) gives

$$t^q(\mathbf{a}_1)|\phi_{\mu,k_1,k_2}\rangle = e^{i2\pi q m_\mu k_1}|\phi_{\mu,k_1,k_2-\chi p}\rangle. \quad (32)$$

Using the boundary condition Eq. (30), the shift $k_2 \rightarrow k_2 - \chi p$ gives

$$|\phi_{\mu,k_1,k_2-\chi p}\rangle = e^{-i2\pi\chi p\sigma_\mu k_1} |\phi_{\mu,k_1,k_2}\rangle. \quad (33)$$

Therefore

$$t^q(\mathbf{a}_1)|\phi_{\mu,k_1,k_2}\rangle = e^{i2\pi(qm_\mu - \chi p\sigma_\mu)k_1} |\phi_{\mu,k_1,k_2}\rangle. \quad (34)$$

But by definition of k_1 , $t^q(\mathbf{a}_1)|\phi_{\mu,k_1,k_2}\rangle = e^{-i2\pi k_1} |\phi_{\mu,k_1,k_2}\rangle$. Thus

$$1 = \chi p\sigma_\mu - qm_\mu. \quad (35)$$

Finally, consider the gap below which the first ν Hofstadter bands are filled. Let

$$t_\nu = \sum_{\mu=1}^{\nu} \sigma_\mu, \quad s_\nu = \sum_{\mu=1}^{\nu} (-m_\mu). \quad (36)$$

Summing Eq. (35) over the occupied bands gives the Diophantine equation $\nu = \chi t_\nu p + s_\nu q$. Here t_ν is the total Chern number of all bands below the ν -th gap, and s_ν is the second integer gap label. The Chern number of the isolated ν -th band is therefore

$$\sigma_\nu = t_\nu - t_{\nu-1}. \quad (37)$$

If several bands below the gap are degenerate or entangled, the same argument applies to the determinant line bundle of the occupied subspace. In that case, σ_μ is replaced by the total Chern number of the occupied subspace, and the phase in Eq. (31) is replaced by the determinant of the corresponding transition matrix.

6 Numerical method for Chern numbers

Suppose at flux $\Phi/\Phi_0 = \chi p/q$ the ν -th eigenstate satisfying $H|\phi_{\nu\mathbf{k}}\rangle = E_{\nu\mathbf{k}}|\phi_{\nu\mathbf{k}}\rangle$ is

$$|\phi_{\nu\mathbf{k}}\rangle = \sum_{n,r} |n, r, \mathbf{k}\rangle P_{nr,\nu}(\mathbf{k}). \quad (38)$$

We define the Bloch functions $|u_{\nu\mathbf{k}}\rangle = e^{-i\mathbf{k}\cdot\mathbf{r}}|\phi_{\nu\mathbf{k}}\rangle$, which satisfy (using $t(\mathbf{a})e^{i\mathbf{k}\cdot\mathbf{r}} = e^{-i\mathbf{k}\cdot\mathbf{a}}e^{i\mathbf{k}\cdot\mathbf{r}}t(\mathbf{a})$)

$$t^q(\mathbf{a}_1)|u_{\nu\mathbf{k}}\rangle = |u_{\nu\mathbf{k}}\rangle, \quad t(\mathbf{a}_2)|u_{\nu\mathbf{k}}\rangle = |u_{\nu\mathbf{k}}\rangle. \quad (39)$$

One should be careful here that $\mathbf{k} = k_1\mathbf{P}_1 + k_2\mathbf{P}_2 = \frac{k_1}{p}\mathbf{g}_1 + k_2\mathbf{g}_2 \in \text{SBZ}$.

The ν -th band Chern number can then be numerically calculated. For multiple bands, we simply replace U below by the determinant. First we discretize $\mathbf{k}_{ij} = (\frac{i}{N_1}, \frac{j}{N_2})$, for $i = 0, 1, \dots, N_1 - 1$, $j = 0, 1, \dots, N_2 - 1$. The U and $\mathcal{F}$ matrices are

$$U_1^\nu(\mathbf{k}_{ij}) = \frac{\langle u_{\nu,\mathbf{k}_{ij}} | u_{\nu,\mathbf{k}_{i+1,j}} \rangle}{|\langle u_{\nu,\mathbf{k}_{ij}} | u_{\nu,\mathbf{k}_{i+1,j}} \rangle|}, \quad U_2^\nu(\mathbf{k}_{ij}) = \frac{\langle u_{\nu,\mathbf{k}_{ij}} | u_{\nu,\mathbf{k}_{i,j+1}} \rangle}{|\langle u_{\nu,\mathbf{k}_{ij}} | u_{\nu,\mathbf{k}_{i,j+1}} \rangle|}, \quad \mathcal{F}^\nu(\mathbf{k}_{ij}) = \ln \left[\frac{U_1^\nu(\mathbf{k}_{ij})U_2^\nu(\mathbf{k}_{i+1,j})}{U_1^\nu(\mathbf{k}_{i,j+1})U_2^\nu(\mathbf{k}_{ij})} \right], \quad (40)$$

where the phase of $\mathcal{F}$ is confined as $[-\pi, \pi)$. The band Chern number reads

$$\sigma_\nu = \frac{1}{2\pi i} \sum_{ij} \mathcal{F}^\nu(\mathbf{k}_{ij}). \quad (41)$$

One should be careful about the inner product of $|u_{\nu\mathbf{k}}\rangle$, which reads $(\mathbf{k} = k_1\mathbf{P}_1 + k_2\mathbf{P}_2 = \frac{k_1}{p}\mathbf{g}_1 + k_2\mathbf{g}_2 \in \text{SBZ})$

$$\langle u_{\nu'\mathbf{k}'} | u_{\nu\mathbf{k}} \rangle = \sum_{n'r'} \sum_{nr} P_{n'r',\nu'}^*(\mathbf{k}') \langle n'r'\mathbf{k}' | e^{i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{r}} | nr\mathbf{k} \rangle P_{nr,\nu}(\mathbf{k}) = [P^\dagger(\mathbf{k}')X(\mathbf{k}',\mathbf{k})P(\mathbf{k})]_{\nu'\nu}, \quad (42)$$

where the X matrix reads (using Eq. (7))

$$\begin{aligned}
[X(\mathbf{k}', \mathbf{k})]_{n'r', nr} &= \langle \psi_{n', \frac{k'_1+r'}{p} \mathbf{g}_1 + k'_2 \mathbf{g}_2} | e^{i[(k'_1 - k_1) \mathbf{g}_1 / p + (k'_2 - k_2) \mathbf{g}_2] \cdot \mathbf{r}} | \psi_{n, \frac{k_1+r}{p} \mathbf{g}_1 + k_2 \mathbf{g}_2} \rangle \\
&= \delta_{\frac{k'_1+r'}{p}, [\frac{k_1+r}{p} + \frac{k'_1-k_1}{p}]} \delta_{k'_2, [k_2+k'_2-k_2]} e^{-i\chi 2\pi \frac{k'_1+r'}{p} [k'_2-k_2-(k'_2-k_2)] - i\chi 2\pi \frac{k'_1-k_1}{p} [k_2+\frac{k'_2-k_2}{2}]} \\
&\quad \times F_{n'n} \left(\frac{(\mathbf{k}' - \mathbf{k})_x + i\chi(\mathbf{k}' - \mathbf{k})_y}{\sqrt{2}} l_B \right) \\
&= \delta_{r'r} e^{-i\chi \frac{\pi}{p} (k'_1 - k_1)(k_2 + k'_2)} F_{n'n} \left(\frac{(\mathbf{k}' - \mathbf{k})_x + i\chi(\mathbf{k}' - \mathbf{k})_y}{\sqrt{2}} l_B \right).
\end{aligned}$$

When a link crosses the SBZ boundary, $\mathbf{k}'$ in $X(\mathbf{k}', \mathbf{k})$ should be treated as the lifted momentum before reduction modulo the SBZ. The eigenvector at the reduced momentum must then be related to the lifted state by the corresponding boundary transition function.